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Duration	10 mins 21 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By using the bubble-up method
- c. By comparing elements sequentially
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Bubble Sort
- c. Binary Search
- d. Merge Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes code execution slower
- b. Provides a basis for other algorithms to work efficiently
- c. Decreases the efficiency of algorithms
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120

b. 80,150,200,470,40,90,300,400,70,120

c. 200,470,80,150,40,90,300,400,70,120

d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion

b. It allows for efficient parallel processing and sorting of data

c. It only works on small datasets

d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **6**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass

b. It is less efficient than bubble sort

c. It is based on the divide-and-conquer approach

d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **7**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It uses the heap data structure
c. It recursively sorts subproblems
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **8**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort
c. Bubble sort
d. Insertion sort

The correct answer is:
Insertion sort

Question **9**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It makes the analysis slower
c. It complicates the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **10**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is difficult to implement
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **11**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It slows down the process
c. It makes it easier to select items based on their relationship to the rest of the items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are split again into smaller sublists
c. They are combined to form a single sorted list
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Selection
c. Distribution
d. Insertion

The correct answer is:
Insertion

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Heap sort
c. Quick sort
d. Merge sort

The correct answer is:
Merge sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Linear Search
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort